

CHAPTER 4

MATHEMATICAL EXPECTATION

4.1 Expected value of a Random Variable

- Definition 4.1- Expected Value

If X is a discrete random variable and $P(x)$ is the value of its probability distribution at x , the expected value of X is

$$E(X) = \sum_X xP_X(x)$$

Correspondingly, if X is a continuous random variable and $f(x)$ is the value of its probability density at x , the expected value of X is

$$E(X) = \int_X xf_X(x)dx$$

In this definition it is assumed that, the sum or the integral exists.
Otherwise, the mathematical expectation is undefined.

Example 4.1.

The probability distribution of X , is given by the following table.

X	0	1	2	3
P(x)	0.4	0.25	0.15	0.2

Find $E(X)$

Example 4.2.

Find $E(X)$ if the pdf of X is given as;

$$f(x) = \begin{cases} k(x+2) & 0 < x < 4 \\ 0 & \text{otherwise} \end{cases}$$

There are many problems which we are interested not only in the expected value of a random variable X , but also in the expected values of random variable related to X .

ex: We might be interested in the random variable Y , whose values are related to those of X by means of the equation $Y = g(x)$.

Theorem 4.1

If X is a discrete random variable and $P(x)$ is the value of its probability distribution at x , the expected value of $g(x)$ is given by

$$E[g(X)] = \sum_X g(x)P_X(x)$$

Correspondingly, if X is a continuous random variable and $f(x)$ is the value of its probability density at x , the expected value of $g(x)$ is

$$E[g(X)] = \int_X g(x)f_X(x)dx$$

Example 4.1 revisited

The probability distribution of X , is given by the following table.

X	0	1	2	3
P(x)	0.4	0.25	0.15	0.2

Find $E[g(X)]$ if $g(x)=5x+2$

Example 4.2 revisited

Find $E(X)$ if the pdf of X is given as;

$$f(x) = \begin{cases} k(x+2) & 0 < x < 4 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[g(X)]$ if $g(x)=3x$

Theorem 4.2

If a and b are constants, then

$$E(aX + b) = aE(X) + b$$

Corollary 4.1. If a is a constant, then $E(aX) = aE(X)$

Corollary 4.2. If c is a constant, $E(c) = c$

Theorem 4.3

If $C_1, C_2, \dots, C_n$ are constants, then

$$E\left[\sum_{i=1}^n c_i g_i(x)\right] = \sum_{i=1}^n c_i E[g_i(x)]$$

Example 4.3

If the probability density of X is given by

$$f(x) \begin{cases} 2(1-x) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that $E[X^r] = 2 / (r+1)(r+2)$

(b) Use above result to evaluate $E[(2X + 1)^2]$

4.2 Moments

Definition 4.2

The moment r^{th} about the origin of a random variable X , denoted by μ'_r

is the expected value of X^r symbolically

$$\mu'_r = E(X^r) = \sum_x x^r \cdot f(x) \text{ for } r = 0, 1, 2, \dots \text{ when } X \text{ is discrete, and}$$
$$\mu'_r = E(X^r) = \int_{-\infty}^{\infty} x^r \cdot f(x) dx \text{ when } X \text{ is continuous.}$$

When $r = 0$,

When $r = 1$,

μ is just expected value of random variable X and in view of its importance in statistics we give it a special symbol (" μ ") and a special name (mean).

Definition 4.3

μ'_1 is called the mean of the distribution of X , or simply the mean of X , and it is denoted by μ .

The special moments we shall define next are of importance in Statistics because they serve to describe the shape of the distribution of a random variable, that is **shape of the graph of its probability distribution or probability density.**

Moments about the mean

Definition 4.4.

The r^{th} moment about the mean of a random variable X , denoted by μ_r is the expected value of $(X - \mu)^r$, symbolically,

$$\mu_r = E[(X - \mu)^r] = \sum_x (x - \mu)^r \cdot f(x) \text{ for } r = 0, 1, 2, \dots \text{ when } X \text{ is discrete}$$

$$\mu_r = E[(X - \mu)^r] = \int_{-\infty}^{\infty} (x - \mu)^r \cdot f(x) dx \text{ when } X \text{ is continuous.}$$

Example: Find μ_0 and μ_1 for any random variable for which μ exists.

The second moment about the mean is of special importance in Statistics because it is indicative of the spread or dispersion of the distribution of a random variable. Thus it is given a special symbol and special name.

Definition 4.5. Variance

$\mu_2 = E(X - \mu)^2$ is called the **variance** of the distribution of X , or simply the variance of X , and it is denoted by

$Var(X)$, $V(X)$ or σ^2 .

The positive square root of the variance , is called **standard deviation**.

Theorem 4.4

$$\sigma^2 = \mu'_2 - \mu^2$$

This is also denoted as computational formula of variance

$$\mathbf{Var}(X) = E(X^2) - [E(X)]^2$$

Example

Theorem 4.5

If X has the variance σ^2 , then

$$\mathbf{Var}(aX + b) = a^2 \mathbf{Var}(X).$$

Proof

4.3 Moment Generating Function

Although the moments of most distributions can be determined directly by evaluating the necessary integral or sums, an alternative procedure sometime provides considerable simplifications. The technique utilizes Moment Generating Functions.

Definition 4.6.

The moment generating function of a random variable X , where it exists is given by,

$$M_X(t) = E[e^{tx}] = \sum_x e^{tx} \cdot f(x) \quad \text{when } X \text{ is discrete}$$

$$M_X(t) = E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx} \cdot f(x) dx \quad \text{when } X \text{ is continuous.}$$

The independent variable is t and we are usually interested in values of t in the neighborhood of zero.

Taylor Series:

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Mclaurin Series:

When $a = 0$ we get this series.

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots = \sum \frac{f^{(n)}(0)}{n!}x^n$$

Exercise. Find the Mclaurine series of e^{tx} .

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots = \sum \frac{f^{(n)}(0)}{n!}x^n$$

Exercise. Find the Mclaurine series of e^{tx} .

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots = \sum \frac{f^{(n)}(0)}{n!}x^n$$

$$e^{tx} = 1 + tx + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3!} + \dots + \frac{t^rx^r}{r!}$$

for discrete case,

$$M_X(t) = E[e^{tx}] = \sum_x e^{tx} \cdot f(x) = \sum_x \left[1 + tx + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3!} + \dots + \frac{t^rx^r}{r!} \right] \cdot f(x)$$

$$\begin{aligned}
&= \sum_x f(x) + \sum_x tx.f(x) + \sum_x \frac{t^2 x^2}{2!} f(x) + \sum_x \frac{t^3 x^3}{3!} f(x) + \dots + \sum_x \frac{t^r x^r}{r!} .f(x) + \dots \\
&= \sum_x f(x) + t \sum_x x.f(x) + \frac{t^2}{2!} \sum_x x^2 f(x) + \frac{t^3}{3!} \sum_x x^3 f(x) + \dots + \frac{t^r}{r!} \sum_x x^r f(x) + \dots \\
&= \mu'_0 + t\mu'_1 + \frac{t^2}{2!}\mu'_2 + \frac{t^3}{3!}\mu'_3 + \dots + \frac{t^r}{r!}\mu'_r + \dots
\end{aligned}$$

It can be seen that in McLaurin's series of the Moment Generating Function of X the coefficient of $t^r/r!$ is μ'_r , the r^{th} moment about the origin.

$$\begin{aligned}
&= \sum_x f(x) + \sum_x tx.f(x) + \sum_x \frac{t^2 x^2}{2!} f(x) + \sum_x \frac{t^3 x^3}{3!} f(x) + \dots + \sum_x \frac{t^r x^r}{r!} .f(x) + \dots \\
&= \sum_x f(x) + t \sum_x x.f(x) + \frac{t^2}{2!} \sum_x x^2 f(x) + \frac{t^3}{3!} \sum_x x^3 f(x) + \dots + \frac{t^r}{r!} \sum_x x^r f(x) + \dots \\
&= \mu'_0 + t\mu'_1 + \frac{t^2}{2!}\mu'_2 + \frac{t^3}{3!}\mu'_3 + \dots + \frac{t^r}{r!}\mu'_r + \dots
\end{aligned}$$

It can be seen that, in McLaurin's series of the Moment Generating Function of X the coefficient of $t^r/r!$ is μ'_r , the r^{th} moment about the origin.

Exercise

Find the moment generating function of the random variable whose probability density is

$$f(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

and use it to find an expression for μ'_r .

The main difficulty of using Maclaurin series of a moment generating function is that of expanding it into a Maclaurin's series. If we are interested only in the first few moments of a random variable, their determination can usually be simplified by using the following theorem.

Theorem 4.6. $\frac{d^r M_X(t)}{dt^r} \Big|_{t=0} = \mu_r'$

Often the work involved in using moment generating function can be simplified by making use of the following theorem.

4.4 Chebyshev's Theorem

Theorem 4.8. (Chebyshev's Theorem)

If μ and σ are the mean and the standard deviation of a random variable X , then for any positive constant k , then probability is at least $\frac{1}{k^2}$ that X will take on a value within k standard deviation of the mean, symbolically,

Example. Find probabilities for 2σ , 3σ , 4σ

Note: The probability given by Chebyshev's Theorem is only a lower bound . In other words it assumes that this probability cannot be less than $1 - 1/k^2$. Only when the distribution of a random variable is known, we calculate the exact probability

Theorem 4.9. (Markov's Inequality)

If X is a random variable with the mean μ for which $f(x)=0$ for x ,

$$Pr(X \geq a) \leq \mu / a$$